

(1) Uniform Random Numbers:

- Use `np.random.rand()` to generate a **1×10** vector of random numbers between **0 and 1**.
- Plot a histogram of the **1×10** vector to visualize the distribution.

(2) Normally Distributed Random Numbers:

- Generate **500** random numbers from a **standard normal distribution** (mean=0, standard deviation=1) using `np.random.normal`.
- Plot a histogram of these numbers to visualize the normal distribution.

(3) Random Integers:

- Use numpy to create a **3×3** matrix of random integers between **1 and 50**.
- Run the code twice:
 - **With a fixed seed (`np.random.seed(10)`)** to ensure reproducibility.
 - **Without setting a seed**, allowing random values to change each time.
- Observe and explain the differences in outputs.

(4) One Sample T test:

- Data: [5.2, 5.4, 5.1, 5.6, 5.3, 5.8]
- Hypothesized mean: 5.5
- Tasks:
 - Calculate the t-statistic and p-value.
 - Interpret and visualize the result at a 0.05 significance level.